Abhinandan

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Experience

Founding Software Engineer — Sept 2025 – Oct 2026
Browzer — *Remote, San Francisco*

- Reduced average LLM request cost by ∼**3.1**× ($0.025 → $0.008) across monitored production traffic through batched agent execution, cache-aware context management, reusable local workflows, and cost-aware model routing.
- Built Browzer's Chrome MV3 recorder and CDP-native browser agent with AX/DOM element capture, cross-iframe support, obstruction checks, and real mouse/keyboard/upload execution.
- Implemented a streaming ReAct runtime across FastAPI + browser extension with SSE tool execution, multi-tab orchestration, safe parallelism, abort/continue semantics, and auditable traces.
- Shipped zero-LLM replay for intact workflows with stateful AI fallback, allowing failed automations and generated content flows to resume from the point of failure instead of restarting.

Founding Software Engineer — Jan 2025 – Jul 2025
Cynos Nexus — *Noida, India*

- Built the AI knowledge base service using FastAPI, LangChain, and vector retrieval pipelines, powering customer-facing search workflows.
- Implemented backend infrastructure for retrieval-augmented AI workflows, including document ingestion, structured data processing, asynchronous workers, Redis-backed queues, and production API serving.
- Productionized AI services with Docker, GitHub Actions, Nginx, cloud deployment, asynchronous processing, and API observability.

Contract Software Engineer — Sept 2024 – Dec 2024
Etkin.ai — *Remote, Turkiye*

- Built a LangChain + pgvector retrieval system; raised recall@5 from 0.71 → 0.82 on a 300-query labeled eval set.
- Developed scalable data-ingestion pipelines using bulk CSV processing and Celery workers, reducing processing time by 40%

Open Source Contributor — June 2024 – Aug 2024
HeroUI — *Y Combinator S24 startup*

- Received a **personal offer from the CEO** to join HeroUI (prev. NextUI) after **16 merged open-source contributions**.

SWE Intern — Jan 2024 – Apr 2024
SkilledUp — *Noida*

- Engineered a context-aware AI code assistant for the core learning dashboard using FastAPI and OpenAI.

Open Source Projects

HiQCache | *SGLang, Quantization* | PR #41365 | GitHub

- Implemented INT8 CPU-tier KV caching in SGLang HiCache with GPU↔CPU compression/restoration, reducing Qwen3-8B host KV memory by **43.75%** (144 → 81 KiB/token) and increasing fixed-budget L2 capacity **1.78**× (54K → 96K tokens).
- Benchmarked on an RTX A6000 using exact-token trace replay (128 requests, 284K prompt tokens), improving cache-hit rate from **57.6%** → **72.0%** against BF16 at equal 8 GB budget, capturing ∼**99%** of the workload's 72.72% oracle reuse ceiling.

RolloutCore | *NCCL, vLLM* | GitHub

- Built a versioned RL rollout control plane around vLLM, orchestrating NCCL hot-weight updates, drain-before-mutate, cache invalidation, and per-trajectory weight provenance without modifying vLLM.
- Benchmarked Qwen3-8B hot updates at **4.25s vs 37.04s** for engine restart (**8.7**× **faster transition**) on 2× RTX 3090s; demonstrated **32 stale KV-cache token hits and divergent generation** without invalidation in an OPT-125M negative-control experiment.

MiniServe | *Python, PyTorch* | GitHub

- Built a from-scratch inference runtime with continuous batching, token-budget scheduling, chunked prefill, recompute preemption, and paged KV with content-addressed prefix caching.
- Implemented a Llama-family path over those paged KV tensors and matched Hugging Face (≤6e-8 float32 logit error; identical greedy tokens); on a 120-request shared-prefix workload, prefix reuse cut computed prefill work by 97%.

Technical Skills

ML: PyTorch, Hugging Face, LoRA / PEFT, Qdrant
Inference: vLLM, SGLang, TensorRT-LLM, NCCL, Quantization
Engineering: Python, C++, TypeScript, FastAPI, Node.js, Redis, PostgreSQL, AWS, GCP, Docker, GitHub Actions

Education & Recognition

Dr. APJ Abdul Kalam Technical University — B.Tech, Computer Science | 2026

International Youth Math Challenge Gold Honour | **Top 1% TypeScript Engineer Globally** (Algora) | **Amazon ML Summer School 2025**